

# UNIT 10 ASSIGNMENT 2

## TASK 1

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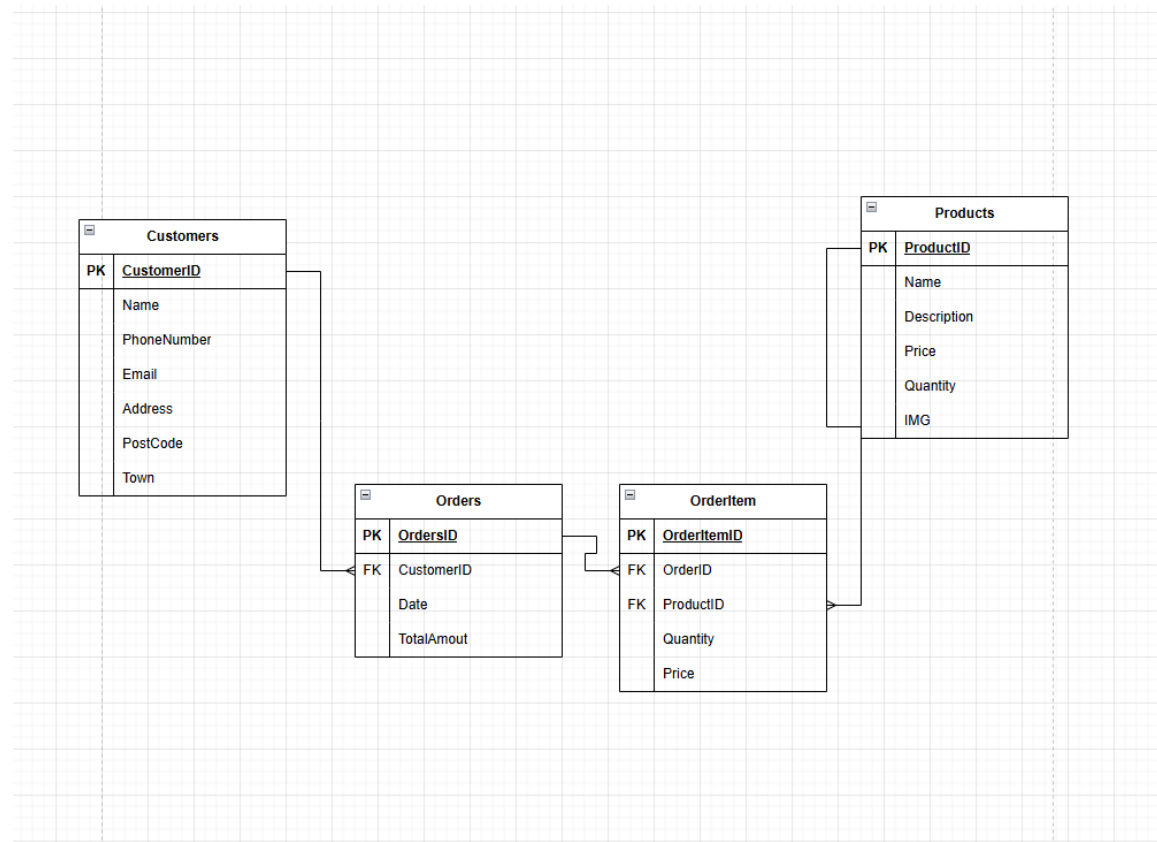
# SYSTEM OVERVIEW

This database is designed for Tech Gear(automated online shop system )that sells technology products such as gaming PCs, graphics cards, keyboards, and accessories. Tech gear is using spreadsheets and paper-based records to manage customer information, stock levels, and orders. This creates problems such as data entry errors, duplicate records, lost paperwork, and incorrect stock levels.

The new automated database system will allow Tech Gear to manage customers, products, orders, and payments efficiently. The system will automatically update stock levels after each purchase, securely process payments, and generate order confirmations.



# ENTITY RELATIONSHIP DIAGRAM (ERD)



# CUSTOMERS

| Field Name  | KEY(PK/FK) | Data Type  | Size/Format  | Required? | Validation Rule                                                                                                                                                                                                                                                   | Error Message                                   |
|-------------|------------|------------|--------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| CustomersID | PK         | Autonumber | Long integer | YES       | None                                                                                                                                                                                                                                                              | None                                            |
| Name        |            | Short text | 100          | YES       | Is not null                                                                                                                                                                                                                                                       | "Name cannot be blank"                          |
| PhoneNumber |            | Short text | 15           | NO        | Like "+[0-9]*"                                                                                                                                                                                                                                                    | Invalid number                                  |
| Email       |            | Short text | 100          | YES       | Like "*@*.*)"                                                                                                                                                                                                                                                     | "Must be a valid email containing an @ symbol." |
| Adress      |            | Short text | 100          | YES       | Is not null                                                                                                                                                                                                                                                       | Address cannot be blank                         |
| PostCode    |            | Short text | 8            | YES       | Like "[A-Z]# [0-9][A-Z][A-Z]"<br>OR Like "[A-Z]## [0-9][A-Z][A-Z]"<br>OR Like "[A-Z][A-Z]# [0-9][A-Z][A-Z]"<br>OR Like "[A-Z][A-Z]## [0-9][A-Z][A-Z]"<br>OR Like "[A-Z]#[A-Z] [0-9][A-Z][A-Z]"<br>OR Like "[A-Z][A-Z]#[A-Z] [0-9][A-Z][A-Z]"<br>OR Like "GIR OAA" | Invalid postcode                                |
| Town        |            | Short text | 100          | YES       | Is not null                                                                                                                                                                                                                                                       | Town cannot be blank                            |

# CUSTOMERS

The Customers table stores information about each customer.

CustomerID is the primary key and uniquely identifies each customer.

It includes details such as name, phone number, email and address.

This table connects to the Orders table using CustomerID. One customer can place many orders.

# PRODUCTS

| Field Name  | KEY(PK/FK) | Data Type  | Size/Format     | Required? | Validation Rule                                                                                                                | Error Message                                              |
|-------------|------------|------------|-----------------|-----------|--------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| ProductID   | PK         | Autonumber | Long integer    | YES       | None                                                                                                                           | None                                                       |
| Name        |            | Short text | 100             | YES       | Is not null                                                                                                                    | "Name cannot be blank"                                     |
| Description |            | Long text  |                 | NO        | None                                                                                                                           | None                                                       |
| Price       |            | Currency   | 2 decimal place | YES       | >=0                                                                                                                            | Price must be greater or equal to 0                        |
| Quantity    |            | Number     | 1 Integer       | YES       | >=0                                                                                                                            | Stock cannot be negative                                   |
| IMG         |            | Hyperlink  | -               | NO        | Like "*.jpg" Or Like "*.jpeg" Or Like "*.png" Or Like "*.gif" Or Like "*.JPG" Or Like "*.JPEG" Or Like "*.PNG" Or Like "*.GIF" | Please enter a valid image file (.jpg, .jpeg, .png, .gif). |

# PRODUCTS

The Products table stores all products available for sale. ProductID is the primary key and identifies each product. It includes name, description, price, quantity and image link. This table connects to the OrderItem table. One product can appear in many orders.

# ORDERS

| Field Name | KEY(PK/FK) | DataType   | Size/Format     | Required? | ValidationRule | Error Message                       |
|------------|------------|------------|-----------------|-----------|----------------|-------------------------------------|
| OrdersID   | PK         | Autonumber | Long integer    | YES       | None           | None                                |
| CustomerID | FK         | Number     | Long Integer    | YES       | Must exist     | Customer does not exist             |
| Data       |            | Data/Time  | Short Date      | YES       | <= Date()      | Order date cannot be in the future  |
| TotalAmout |            | Currency   | 2 decimal place | YES       | >=0            | Price must be greater or equal to 0 |



# ORDERS

The Orders table stores information about each order placed.

OrdersID is the primary key. CustomerID is a foreign key linking to the Customers table.

It includes the order date and total amount.

One order can contain multiple products.

# ORDERITEM

| Field Name  | KEY(PK/FK) | Data Type  | Size/Format      | Required? | Validation Rule | Error Message                   |
|-------------|------------|------------|------------------|-----------|-----------------|---------------------------------|
| OrderItemID | PK         | Autonumber | Long integer     | YES       | None            | None                            |
| OrderID     | FK         | Number     | Long Integer     | YES       | Must exist      | Order does not exist            |
| ProductID   | FK         | Number     | Long Integer     | YES       | Must exist      | Product does not exist          |
| Quantity    |            | Number     | Integer          | YES       | >0              | Quantity must be greater than 0 |
| Price       |            | Currency   | 2 decimal places | YES       | >=0             | Price must be 0 or greater      |

# ORDERITEM

The OrderItem table links Orders and Products together.  
It contains OrderID and ProductID as foreign keys.  
Each record represents one product inside an order.  
This creates a many-to-many relationship between orders and products.